Santiago Monrroy López

Software Engineer | Backend, APIs & Quality Engineering
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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience across the software development lifecycle (SDLC), specializing in backend API development, Clean Architecture (DDD), and AI-assisted engineering workflows. Combines hands-on development expertise in PHP (Laravel), Java (Spring Boot/Hibernate), TypeScript, Node.js, Vue.js/React, and relational SQL with an engineering mindset for unit testing (TDD), E2E test automation, root cause analysis, and production reliability. Experienced in leveraging local LLMs, structured prompting, and agent-driven workflows to accelerate requirements specification, automated test generation, and continuous software delivery. Passionate about technical mentoring, team coordination, and cross-functional collaboration in Agile environments.

WORK EXPERIENCE

Quality Analyst (Production Reliability & Live Operations) | Electronic Arts

Nov 2024 - May 2026

- **Performance, Stability & Release Phase Monitoring:** Monitored system stability, millisecond latency metrics, and response times within continuous deployment pipelines across stabilization (regression) and final release phases; audited server logs, cross-platform console compatibility, and wallet balance consistency across in-game store purchases, skin rendering, and microtransactions during high-concurrency staging playtests.
- **Production Reliability & Root Cause Analysis (RCA):** Investigated critical defects and mitigated live operational risks for AAA titles under continuous deployment, conducting technical Root Cause Analysis (RCA) across server logs, latency spikes, and backend data consistency to prevent production regressions ahead of official launches.
- **Process Optimization, Coordination & Onboarding Velocity:** Reduced team onboarding and technical training ramp-up times by 83% (from 3 months down to 2 weeks) by authoring centralized engineering documentation, technical runbooks, and video guides. Coordinated test suite allocation and prioritization tailored to each team member's technical strengths while enforcing rigorous test specification standards.
- **Test Server Provisioning, DevKits & Technical Triage:** Configured and provisioned dedicated test servers (adjusting runtime parameters and network ports) to support simultaneous multiplayer test connections, configured proprietary console **DevKit** software across platforms, and explored operational workflow automation (n8n-style). Led technical triage sessions in Jira and engineered advanced **JQL** filters and analytical dashboards in **QMetry** (following the migration from **TestRail**) to track delivery health in real time.

Technical Support Analyst | Teleperformance (Expedia Account)

Mar 2024 - Oct 2024

- **Systems Troubleshooting, SLAs & Escalation:** Diagnosed operational failures, asset loading defects, and API/CRM integration discrepancies within global hospitality PMS platforms under strict **SLA** compliance; structured actionable English technical escalation packages (steps to reproduce, server logs, HTTP error payloads, and transaction IDs) in **Salesforce**, accurately routing tickets to the corresponding engineering teams to accelerate bug resolution.
- **Transactional Integrity & Financial Audit:** Audited financial workflows, payment reversions, and refund processing across 4–5 interconnected internal platforms, verifying database state consistency and business invariant adherence across multi-platform operations.

Web Developer | LaLicorera.com

Aug 2022 - Aug 2023

- **Backend API Development, Clean Architecture & Unit Testing (TDD):** Architected, refactored, and implemented decoupled RESTful endpoints in PHP (**Laravel**) under **Clean Architecture (DDD)** across Controllers, Services, Repositories, and Entities, advocating for **DTO** patterns; implemented **unit tests under a TDD methodology** to validate business rules and accurate state updates on domain objects.
- **API Integration, Documentation & Concurrency Testing:** Built, executed, and documented comprehensive functional integration test suites in **Postman** for every developed module, validating JSON schemas, HTTP status codes, payloads, and conducting concurrency and idempotency testing against repeated simultaneous requests.

- **Vue.js Frontend Refactoring, Web Performance & New Features:** Rapidly transitioned from React to **Vue.js** (leveraging Redux/Vuex-style state management, UI component frameworks, and component lifecycle hooks), refactoring large monolithic files (+500 lines) into reusable modular components. Optimized e-commerce page load performance by transforming image assets into **WebP** format, and developed end-to-end into production the commercial **custom laser bottle engraving** feature.
- **Data Persistence & Relational SQL Validation:** Directly verified transactional data persistence, relational integrity, and database records using **MySQL Workbench** and **phpMyAdmin**, supporting internal administrative systems for inventory control and the accounting product pricing application.

Software Developer Trainee | Globant
Mar 2019 - Sep 2019

- **Backend Development with Java, Spring Boot & Hibernate (ORM):** Developed and structured enterprise backend services using **Java and Spring Boot** for an internal sports event management application, applying annotation-driven configuration, layered architecture, and Object-Relational Mapping (**ORM with Hibernate**) connected to **MySQL** databases. Designed and implemented role-specific **DTOs (Data Transfer Objects)** to decouple and secure data exposure while managing input validation and null safety.
- **React Frontend Development & Engineering Foundations:** Built and integrated user interface components in **React with JavaScript** consuming internal REST APIs, solidifying software engineering best practices, intensive English technical documentation reading, and collaborative workflows.

Technical Initiatives & AI Engineering Projects | *Independent*
Currently

- **Testbed Architecture & Unit Testing:** Designed and built an independent backend testbed in **Node.js** and **TypeScript** structured under Clean Architecture; implemented isolated unit test suites using **Vitest** and repository mocks to validate business invariants and use cases before End-to-End automation with Playwright.
- **E2E Test Automation Architecture (Playwright & TypeScript):** Designing scalable test automation suites leveraging **Playwright with TypeScript**, implementing the **Page Object Model (POM)** pattern to separate element locators and page actions from test scripts for improved maintainability. Preparing API testing workflows using **Postman** for REST endpoint and payload validation.
- **Local LLM Environment & Modelfile Customization:** Configured and deployed local AI environments using **Ollama** and **OpenCode**. Customized model behaviors, context windows, resource allocation, and parameter tuning (e.g., temperature optimization for deterministic outputs) via custom **Modelfiles** to support secure, offline software engineering and QA operations.
- **AI-Assisted Engineering Operations & Prompt Engineering:** Leveraged structured Prompt Engineering frameworks (**RTF**, **CREATE**) alongside **Agent Skills** and **Model Context Protocol (MCP)** to integrate local LLMs with workspace file I/O, automating edge-case test suite generation, synthetic JSON mocks, and code review assistance for automation scripts.

EDUCATION

- **Bachelor's Degree in Systems and Computer Engineering** (In Progress)
  Universidad Central
- **Technologist Degree in Systems Development** (Graduated 2022)
  SENA – Servicio Nacional de Aprendizaje
- **Technical Degree in Software Programming** (Graduated 2019)
  SENA – Servicio Nacional de Aprendizaje

CERTIFICATIONS & SKILLS

Certifications

- Programming with Emphasis on Programming Skills – Universidad de Antioquia (800 hrs)
- Programming with Emphasis on Mobile App Development – Universidad de Antioquia (800 hrs)
- Design UX/UI – Universidad Nacional (400 hrs)
- ISTQB Certified Tester Foundation Level (CTFL v4.0) Candidate

Skills & Technologies

- **Backend, Frontend & Languages:** TypeScript, Node.js, PHP (Laravel), Java (Spring Boot, Hibernate ORM), Vue.js (Vuex), React, JavaScript, SQL (MySQL Workbench, phpMyAdmin, PostgreSQL), RESTful APIs, DTOs, Clean Architecture / DDD, TDD, Git.
- **DevOps, Infrastructure & Tooling:** Docker (Containerization), CI/CD (GitHub Actions, Jenkins), Linux/Unix environment, Test Server Provisioning & DevKits, n8n.
- **Testing & Reliability:** Vitest (Unit Testing & Mocks), Playwright (E2E & POM), Postman (API Contract Validation & Documentation), Root Cause Analysis (RCA), Live Operations, JIRA, QMetry, TestRail, Salesforce.
- **AI & Engineering Productivity:** Ollama, OpenCode, Model Context Protocol (MCP), Local LLMs, Modelfiles, Agent Skills, Prompt Engineering (RTF, CREATE).

Languages

- **English:** Fluent / CEFR B2 Certified - British Council
- **Spanish:** Native

Soft Skills

Attention to detail, critical thinking, team coordination, technical mentoring, defect triage, data validation, cross-functional collaboration, knowledge sharing.